GDG FUNAAB

Introduction to AI & Machine Learning

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What is AI?

Machines performing **human-like tasks**

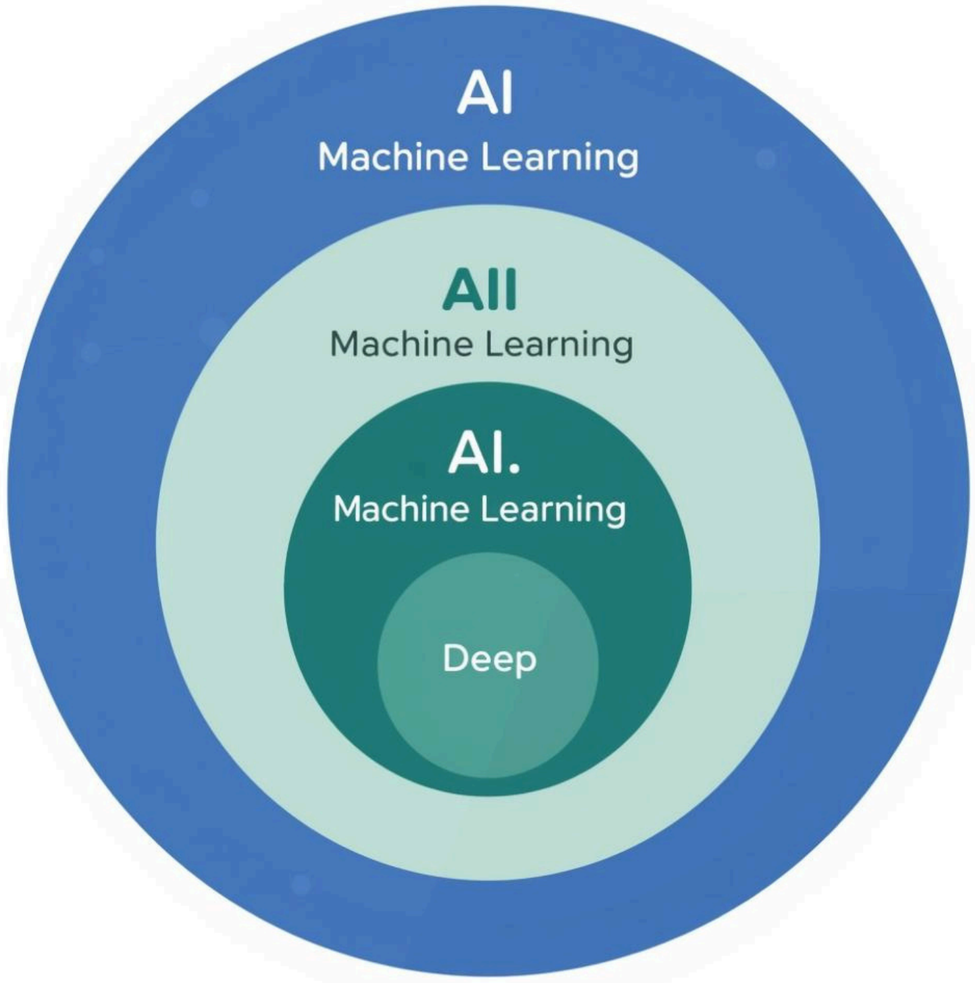
What is Machine Learning?

Systems that learn patterns

AI, ML, and Deep Learning

Understanding their relationships and differences

Artificial Intelligence (AI) encompasses the broader concept of machines performing tasks usually requiring human intelligence, while Machine Learning (ML) is a subset of AI that uses data patterns for learning. Deep Learning is a further specialization within ML.



Why Machine Learning Matters

Real-world applications impacting our daily lives

Healthcare

Machine learning enhances **disease prediction** accuracy, enabling early interventions and improving patient outcomes through data-driven insights and analysis.

Finance

In finance, ML algorithms analyze transaction patterns to detect **fraud** quickly, protecting consumers and institutions from financial losses.

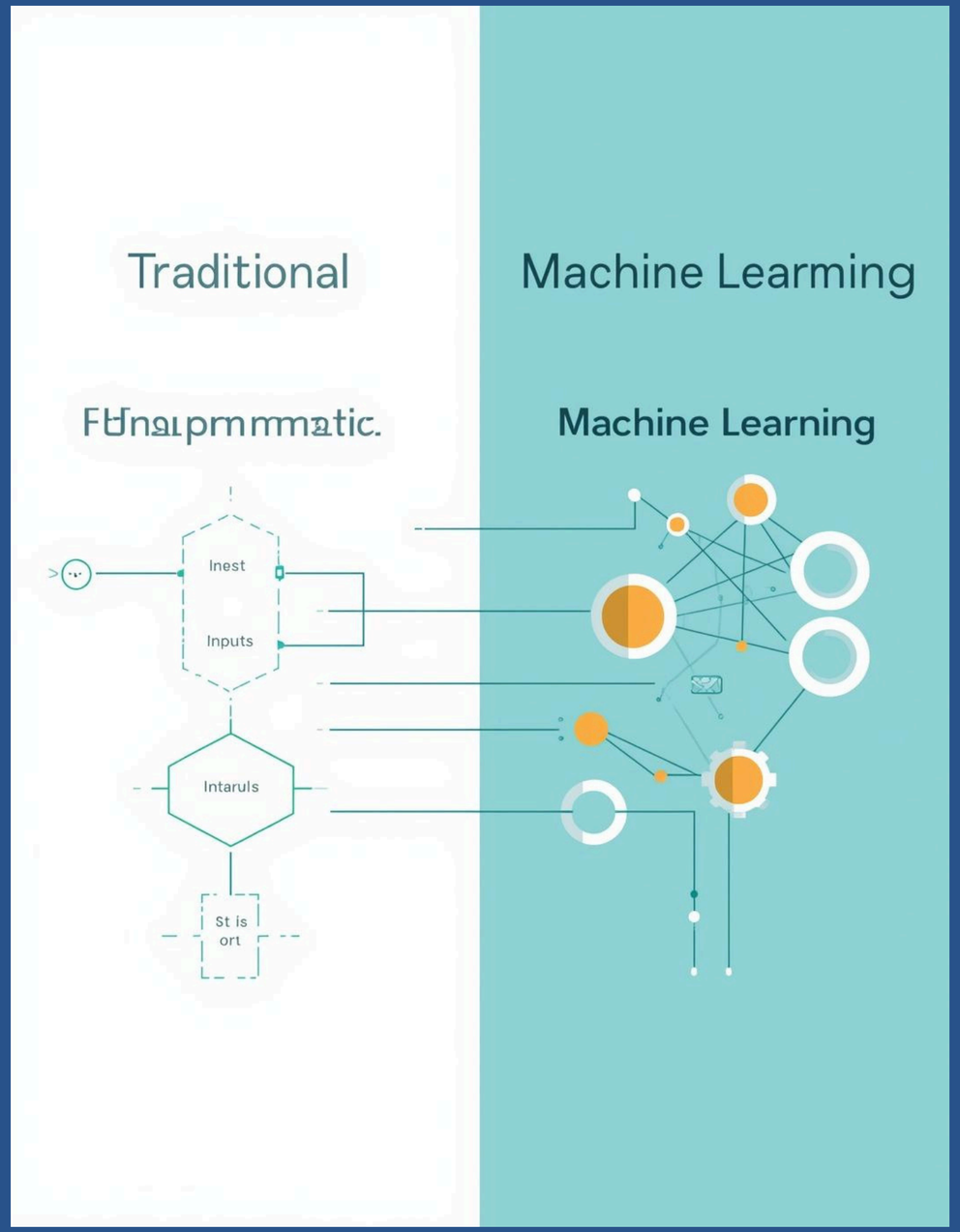
Social Media

Social media platforms utilize machine learning for **recommendations**, personalizing user experiences based on individual preferences and behavior patterns.

Traditional Programming vs ML

Comparing Two Approaches

Traditional programming relies on explicit rules, while machine learning uses input data to learn patterns, adapting its responses without needing predetermined instructions or rules.



Types of Machine Learning

Understanding the three main learning approaches in ML

Supervised Learning

In supervised learning, models learn from labeled data, allowing them to make predictions based on input-output pairs provided during training.

Unsupervised Learning

Unsupervised learning involves training models on unlabelled data, enabling them to identify patterns and group similar data points without explicit guidance.

Reinforcement Learning

Reinforcement learning teaches models through trial and error, where they learn to make decisions based on rewards or punishments received from actions.

Key Terminology

Essential terms for understanding machine learning concepts

Dataset

A **dataset** is a collection of data that serves as the foundation for training a machine learning model, crucial for accuracy.

Features

Features are the input variables or characteristics in a dataset that help the model learn and make predictions effectively.

Labels

Labels are the correct answers or outputs associated with the features in a dataset, guiding the model during the learning process.

The Machine Learning Workflow

Understanding the core ML process flow

The machine learning workflow consists of four essential steps: collect data, train the model, evaluate its performance, and make predictions. This iterative process is fundamental to successful machine learning practices.

Data Cleaning Essentials

Handling Missing Values

Missing values can lead to inaccurate predictions. Identifying and addressing these gaps is crucial for effective data analysis and training.

Removing Duplicates

Duplicate records can skew results, leading to misleading insights. Regularly checking for and eliminating duplicates ensures data integrity.

Fixing Inconsistencies

Inconsistent data formats can disrupt model performance. Standardizing formats enhances clarity, making data more reliable for analysis and processing.

Dealing with Outliers

Outliers can distort results and mislead interpretations. Identifying and handling them appropriately is vital for creating robust machine learning models.

Practical Example

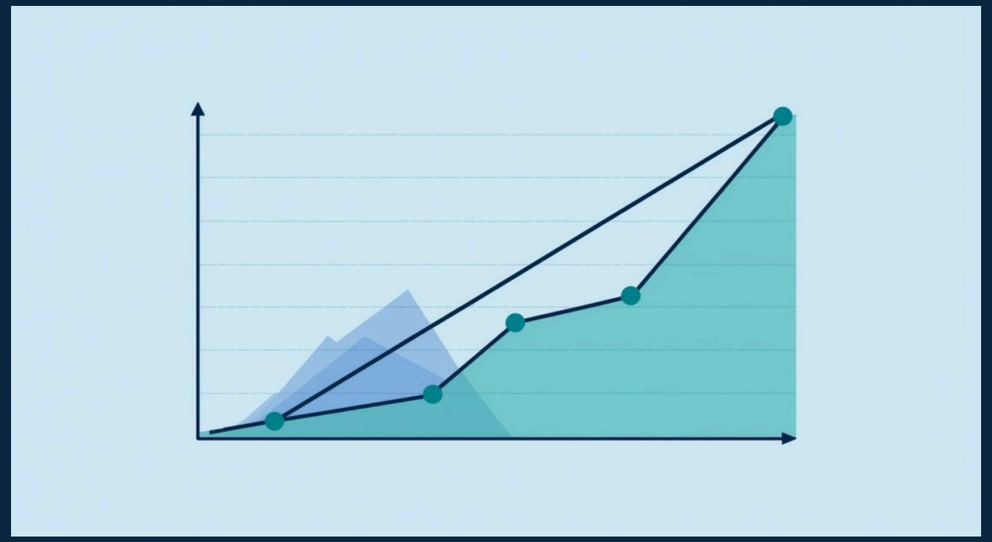
Predicting House Prices

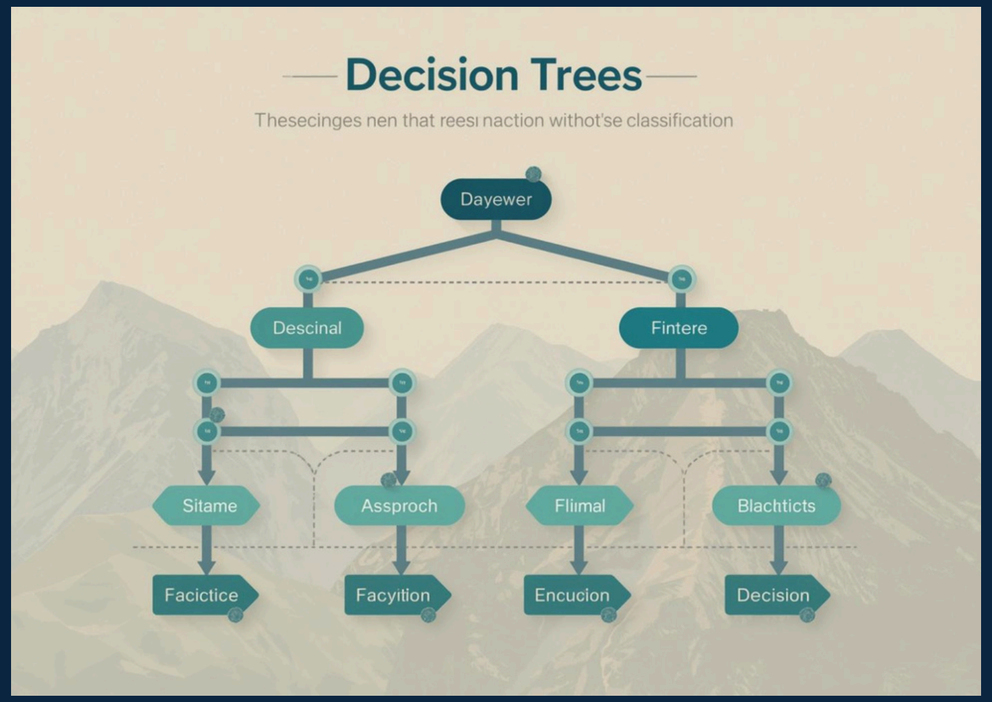
In this example, we explore how **machine learning algorithms** utilize historical data like size and location to predict house prices, helping buyers make informed decisions.

Common ML Algorithms

Understanding Key Techniques

Linear regression and decision trees are foundational algorithms in machine learning, used for predicting outcomes based on data patterns and making informed decisions.





Common Challenges

Understanding key pitfalls in machine learning practices

Overfitting

Overfitting occurs when a model learns noise instead of patterns, resulting in poor performance on unseen data and decreased generalization.

Bias

Bias in machine learning arises from using flawed datasets, leading to unfair and inaccurate predictions that can perpetuate existing inequalities.

Poor Data Quality

Poor data quality can significantly hinder the effectiveness of machine learning models, causing inaccuracies and unreliable predictions that affect decision-making.

Essential Skills

Key tools and knowledge for Machine Learning beginners

Python Programming

Python is a versatile language widely used in ML, allowing you to implement algorithms and manage data efficiently.

Basic Math

A foundational understanding of statistics helps you interpret data and results, making it easier to draw meaningful conclusions.

Key Libraries

Familiarity with libraries like NumPy, Pandas, and scikit-learn streamlines the development process and simplifies complex tasks in ML.

Roadmap to Mastery

Learn Python Basics

Understanding Python is essential for implementing machine learning algorithms and manipulating data effectively in your projects.

Master Data Handling

Data manipulation using libraries like Pandas is crucial for preparing datasets and ensuring quality inputs for your models.

Understand ML Fundamentals

Grasping the core principles of machine learning will provide a solid foundation for further exploration and advanced techniques.

Build Projects

Real-world projects will solidify your knowledge and skills, allowing you to apply concepts and gain practical experience in ML.

Any Questions?

Let's discuss AI and Machine Learning together